

SYSTEM ANALYSIS AND DESIGN

Unit-I: Overview of System Analysis and Design

Comprehensive Study Notes

1. Introduction to System

1.1 What is a System?

A **system** is a set of interconnected and interdependent components that work together to achieve a common objective.

Definition (James O'Brien): "A system is a set of elements that are connected and work together to achieve a specific goal."

1.2 Characteristics of a System

Characteristic	Description
Components	Made up of interconnected parts
Interaction	Components interact with each other
Boundary	Has a boundary that separates it from environment
Purpose	Designed to achieve specific goals
Environment	Exists within a larger environment
Input	Receives inputs from environment
Output	Produces outputs to environment
Feedback	Uses feedback for control and adjustment

1.3 Types of Systems

Open vs Closed Systems

+-----+	+-----+
OPEN SYSTEMS	CLOSED SYSTEMS
+-----+	+-----+
Interacts with	No interaction with
environment	environment
Business organizations	Limited to theory
are open systems	
+-----+	+-----+

Other Classifications

Classification	Type	Description
Based on Interaction	Open System	Interacts freely with the environment
	Closed System	No interaction with the environment
Based on Nature	Physical System	Tangible, concrete systems
	Abstract System	Conceptual, theoretical systems
Based on Behavior	Deterministic System	Predictable, fixed behavior
	Probabilistic System	Uncertain, probability-based behavior

2. Information System

2.1 What is an Information System?

An **Information System** is a system that collects, processes, stores, and distributes information to support decision-making and control.

2.2 Business Information Systems

Type	Purpose
Transaction Processing Systems (TPS)	Process routine transactions
Management Information Systems (MIS)	Provide management reports
Decision Support Systems (DSS)	Support decision-making
Executive Information Systems (EIS)	Support executive decisions
Expert Systems	Mimic human expertise

3. System Development Life Cycle (SDLC)

3.1 What is SDLC?

System Development Life Cycle (SDLC) is a systematic approach to developing information systems. It provides a structured framework for planning, building, and maintaining systems.

3.2 Phases of SDLC

Phase	Description	Activities
Planning Phase	Identify the need for a new system	Project selection, feasibility study
Analysis Phase	Understand current system and requirements	Requirement gathering, analysis
Design Phase	Design the new system	System design, interface design
Implementation Phase	Build and deploy the system	Coding, testing, deployment
Maintenance Phase	Maintain and improve the system	Bug fixes, enhancements

3.3 SDLC Models

+-----+	+-----+
WATERFALL MODEL	AGILE MODEL
+-----+	+-----+
Sequential	Iterative
Linear	Flexible
Rigid	Customer-centric
+-----+	+-----+

4. Feasibility Analysis

4.1 What is Feasibility Analysis?

Feasibility Analysis is the process of evaluating whether a proposed system is viable and worthwhile.

4.2 Types of Feasibility

Type	Description
Technical Feasibility	Is the technology available?
Economic Feasibility	Is it financially viable?
Operational Feasibility	Will it work in the organization?
Schedule Feasibility	Can it be completed on time?
Legal Feasibility	Does it comply with laws?
Behavioral Feasibility	Will users accept it?

5. Project Selection

5.1 What is Project Selection?

Project Selection is the process of evaluating and choosing which projects to undertake based on their value, feasibility, and alignment with organizational goals.

5.2 Criteria for Project Selection

Criteria	Description
Strategic Alignment	Aligns with business strategy
Cost-Benefit Analysis	Benefits outweigh costs
Feasibility	Technically and operationally feasible
Risk	Acceptable level of risk
Urgency	Urgency of the need
Resources	Resources available
ROI	Return on investment

6. System Design & Testing

6.1 System Design

System Design is the process of creating a blueprint for the new system.

Design Activities

Activity	Description
Architecture Design	Overall structure and components
Interface Design	User interface design
Database Design	Data storage structure
Program Design	Detailed program specifications
Security Design	Security measures

Limitations

Limitation	Description
Technical Limitations	Technology constraints
Budget Constraints	Financial limitations
Time Constraints	Schedule limitations
Resource Limitations	Human and other resources
Scope Creep	Expanding requirements

6.2 Testing

Testing is the process of verifying that the system works correctly.

6.3 Evaluation

Evaluation is the process of assessing whether the system meets its objectives.

Criteria	Description
Effectiveness	Does it achieve goals?
Efficiency	Does it use resources well?
User Satisfaction	Are users satisfied?
Reliability	Is it reliable?
Maintainability	Is it easy to maintain?

7. System Analyst

7.1 Sources of Requests

Source	Description
Users	Requests from system users
Management	Requests from management
IT Department	Technical requests
External Sources	Legal requirements, competition
Audit Reports	Audit findings

7.2 Qualities of a System Analyst

Quality	Description
Technical Knowledge	Understanding of technology
Analytical Skills	Problem-solving ability
Communication Skills	Written and verbal communication
Interpersonal Skills	Working with people
Management Skills	Project management
Business Knowledge	Understanding of business
Creativity	Innovative thinking
Listening Skills	Active listening
Ethics	Professional ethics

7.3 Requirement Gathering Techniques

Questionnaires

- **Types:** Open-ended, closed-ended, mixed
- **Designing:** Clear objectives, simple language, logical flow, unbiased questions, pilot testing

Interviews

- **Types:** Structured, unstructured, semi-structured
- **Preparation:** Define objectives, research background, prepare questions, schedule, set agenda

Observation

Directly observing users performing their tasks to understand current processes and requirements.

Record Review

Analyzing existing documents, reports, and records to gather information about the current system.

8. Questions

Short Answer Questions (2-5 Marks)

1. What is a system?
2. What are the characteristics of a system?
3. What is SDLC?
4. What are the phases of SDLC?
5. What is feasibility analysis?
6. What is project selection?
7. What are the qualities of a system analyst?
8. What is the role of a system analyst?
9. What are the types of systems?
10. What is an information system?

Long Answer Questions (10 Marks)

1. Explain the concept of a system with its characteristics.
2. Explain the System Development Life Cycle.
3. Explain the types of feasibility analysis.
4. Explain the qualities of a system analyst.
5. Explain the different SDLC models.

Objective Questions (1 Mark)

1. A system is: (a) A collection of components (b) A single component (c) A product (d) A service
2. SDLC stands for: (a) System Development Life Cycle (b) Software Design Life Cycle (c) System Design Life Cycle (d) None
3. The first phase of SDLC is: (a) Design (b) Analysis (c) Planning (d) Implementation
4. Feasibility analysis is done in: (a) Planning phase (b) Design phase (c) Implementation phase (d) Testing phase

Quick Revision

System Characteristics

System

- | -- Components
- | -- Interaction
- | -- Boundary
- | -- Purpose
- | -- Environment
- | -- Input
- | -- Output
- | -- Feedback

SDLC Phases

SDLC

- | -- Planning
- | -- Analysis
- | -- Design
- | -- Implementation
- | -- Maintenance

Feasibility Types

Feasibility

- | -- Technical
- | -- Economic
- | -- Operational
- | -- Schedule
- | -- Legal
- | -- Behavioral

End of Unit-I Notes